Quadrilaterals - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

A quadrilateral has diagonals that are equal in length, bisect each other and cross

What is the quadrilateral?

Equal diagonals: rectangle or square

Crossing at 90° as well: square

Answer: Square

Question 2 *Practice question written for this worksheet (not from a past paper)*

Opposite angles of a parallelogram are $(3x + 10)^\circ$ and $(5x - 20)^\circ$.

Work out x.

Opposite angles are equal

$3x + 10 = 5x - 20$

Answer: x = 15

Question 3 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2FR · June 2022 · Question 13*

Question 7 June 2022 · Paper 2FR · Q13 — **2 marks**

Part	Working / answer	Mark	Notes
	½ × (6 + 13) × 3	M1	
	28.5 m²	A1	

Question 4 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · November 2025 · Question 15*

Question 2 November 2025 · Paper 2F · Q15 — **2 marks**

Part	Working / answer	Mark	Notes
	½ × (14 + 29) × 8	M1	
	172 cm²	A1	

A+ Maths Hub International Community Schools

Question 5

Practice question written for this worksheet (not from a past paper)

$ABCD$ is a rhombus. Angle $ABC = (2x + 10)^\circ$ and angle $BCD = (3x - 20)^\circ$.

Work out the size of angle ABC.

Adjacent angles add to 180°

$5x - 10 = 180$, $x = 38$

$2 \times 38 + 10$

Answer: 86°

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 15

15	$\frac{1}{2} \times 6 \times 11 \times \sin 118 (= 29.1...)$		3	M1	for the area of half of the kite
	eg $2 \times \frac{1}{2} \times 6 \times 11 \times \sin 118$			M1	for a complete method
		58.3		A1	accept 58.2 – 58.3
					Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 15

15	$\frac{1}{2} \times 8.9 \times 6.7 \times \sin 74°$ or 28.6(600...)	$h = 6.7 \sin 74°$ or 6.44(0...)		3	M1	or a complete correct method to find the perpendicular height
	$\times 2$	8.9×"6.44(0...)"			M1	(dep) for a complete method to find area of parallelogram
			57.3		A1	for answer rounding to 57.3 (57.320...)
						Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 19

19	$0.5 \times 6.4 \times 9.7 \times \sin 110$ (= 29.16…)			M1	M2 for $6.4 \times 9.7 \times \sin 110$
	$2 \times$ "29.16…"			M1	
		58.3	3	A1	for 58.3 – 58.4
	alternative				
	$AC = \sqrt{6.4^2 + 9.7^2 - 2 \times 9.7 \times 6.4 \times \cos 110}$ (=13.323...) $DAC = \sin^{-1}(\frac{\sin 110}{'13.323'} \times 9.7)(= 43.167...)$ or $ACD = \sin^{-1}(\frac{\sin 110}{'13.323'} \times 6.4)(= 26.83...)$			M1	For method to find AC and angle DAC or angle ACD
	Area = (sin '43.167..' × 6.4 × 2 × '13.323..') ÷ 2 Or area = (sin '26.83..' × 9.7 × 2 × '13.323...') ÷ 2			M1	find DB and then area using half product of diagonals
		58.3		A1	for 58.3 – 58.4
					Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2024 · Question 7

Question 10 November 2024 · Paper 1F · Q7 — **4 marks**

Part	Working / answer	Mark	Notes
	Angle ADC = 180 − 138 = 42°	M1	
	x = 360 − 125 − 107 − 42	M1	
	x = 86	A1	
	Reasons: angles on a straight line add up to 180°; angles in a quadrilateral add up to 360°	B1	Both reasons needed

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2FR · January 2022 · Question 9

Question 16 January 2022 · Paper 2FR · Q9 — **4 marks**

Part	Working / answer	Mark	Notes
(i)	360 − 92 − 44 = 224	M1	
	224 ÷ 2	M1	
	x = 112	A1	
(ii)	Angles in a quadrilateral add up to 360°	B1	

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 13

Question 23 January 2022 · Paper 1F · Q13 — **4 marks**

Part	Working / answer	Mark	Notes
	Area = ½ × (2.5 + 3.5) × 6.5 = 19.5 m²	M1	
	Two coats: 2 × 19.5 = 39 m²	M1	
	39 ÷ 12 = 3.25	M1	
	4 tins	A1	Must round up

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 3

3	$(4x - 27) + (3x + 46) = 180$ oe or "expression for C" + $(3x + 10) = 180$ **or** $7x + 19 = 180$ **or** $3x + 46 + 4x - 27 + 3x + 10 + [\text{"}180 - (3x + 10)\text{"}] = 360$		4	M1	Sum angles *A* and *B* to 180, or find an expression for *BCD* and sum all angles to 360. [condone missing brackets and condone use of any letter or expression for angle *C* (even *x* or *BCD*)]
				A1	$x = 23$
	eg $3 \times$"23" + 46 (= 115) **or** eg $180 - (3 \times$"23" + 10) (= 101)			M1ft	dep on M1 using **their** ***x*** to calculate a value for angle *B* or 'their' *C* (cannot be a negative value and cannot just be *x*)
	Correct answer scores full marks (unless from obvious incorrect working)	115		A1	Allow $3x + 46$ or *ABC* if 115 is clearly seen in working or on diagram
					Total 4 marks

A+ Maths Hub International Community Schools

Quadrilaterals - Mark scheme

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 4*

Question	Working	Answer	Mark	Notes
4	eg $5x-1=3x+7.4$ oe **or** eg $10x-2+48$ **or** $6x+14.8+48$ **or** $24+24+5x-1+3x+7.4$ oe		4	M1 a correct equation to find x **or** a correct expression for the perimeter in terms of x
	$x=4.2$			A1 the correct value of x (implies previous mark)
	$2\times24+2(5\times$ "4.2" $-1)$ oe **or** $2\times24+2(3\times$ "4.2" $+7.4)$ oe **or** $2\times24+(5\times4.2-1)+(3\times4.2+7.4)$ oe eg $24+24+20+20$ oe			M1dep on a correct method to find the perimeter – use of positive x from correct working (1st M1 awarded for an equation) **and** only if used the same measurement for AD and BC
	working required	88		A1 cao dep on either M1 or $x=4.2$
				Total 4 marks

Question 14 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 17*

Question 17 June 2026 · Paper 1F · Q17 — **4 marks**

Part	Working / answer	Mark	Notes
	h = 175 ÷ 25 = 7	M1	
	Trapezium = ½ × (11 + 25) × 7 = 126	M1	
	126 ÷ 175 × 100	M1	
	72%	A1	

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 3*

3	$ADC = 180-58$ (= 122) **or** $EDF = 122$ **or** $CDE = 58$ **or** $ADF = 58$			M1 may be seen marked on the diagram
	e.g. $DEF = 58\div2$ **or** $DEF = (180-122)\div2$			M1 complete method to find angle DEF
		29		A1
			5	B2 dep on M2 for fully correct reasons for their method (B1 dep on M1 for one correct reason stated and used) e.g. Allied angles, co-interior angles, Alternate angles, Corresponding angles, Vertically opposite angles are equal (or Vertically opposite angles are equal), Angles on a straight line add up to 180°(or angles on a straight line add to 180°), Sum of two angles in a triangle are equal to opposite exterior angle, Angles in a triangle add up to 180°(or Angles in a triangle add up to 180°), Base angles in an isosceles triangle Angles in a quadrilateral add up to 360. (accept "4-sided shape"or parallelogram) Opposite angles of a parallelogram are equal
				Total 5 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2011 · Question 6*

6. (a)	½ (6+8)x5 or ½ x2x5 + 6x5	35	2	M1 A1
(b)	8 – 6 (=2) and 5 seen $(PQ^2=)$ $(\text{"}8-6\text{"})^2+5^2$ (=29) (PQ=) √"29"	5.39	4	B1 could be seen on diagram M1 (dep) $(\theta=)\ \tan^{-1}(5/\text{"}8-6\text{"})$ (=68.2 or better) M1 (dep) (PQ=) "8–6"/ cos "68.2" or 5 / sin "68.2" A1 5.38516..... awrt 5.39
				Total 6 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 8

8	$\frac{14+AB}{2}\times 15 = 360$ oe **or** $360-14\times 15 (=150)$ oe		6	M1	for setting up an equation using the area of the trapezium **or** method to find the area of the triangle
	$AB = 34$ **or** $MB = 20$ (where M is point on AB such that MC is perpendicular to AB)			A1	could be seen on diagram
	$(CB^2 =) 15^2 + 20^2 (= 625)$ **or** $(CB^2 =) 15^2 + MB^2$			M1	allow use of their MB
	$(CB =) \sqrt{15^2 + 20^2} (= 25)$ **or** $(CB =) \sqrt{15^2 + MB^2}$			M1	allow use of their MB
	14 + 15 + "34" + "25" oe **or** 14 + 15 + 14 + MB + CB oe			M1ft	(dep on previous two M marks) for a method to find the perimeter of the trapezium, allow use of their MB and CB
	Correct answer scores full marks (unless from obvious incorrect working)	88		A1	cao
					Total 6 marks

A+ Maths Hub International Community Schools

CHALLENGE QUESTIONS

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2012 · Question 9*

9.	(a)	Complete, correct expression which, if correctly evaluated, gives 48 eg $4\times\frac{1}{2}\times6\times4$, $2\times\frac{1}{2}\times12\times4$, $\frac{1}{2}\times12\times8$		3	M2	M1 for correct expression for area of one relevant triangle eg $\frac{1}{2}\times6\times4$, $\frac{1}{2}\times6\times4\sin90°$, $\frac{1}{2}\times8\times6$, $\frac{1}{2}\times12\times4$
			48		A1	cao
	(b)	$4^2+6^2=16+36=52$		3	M1	for squaring and adding
		$\sqrt{4^2+6^2}$			M1	(dep) for square root
			7.21		A1	for answer which rounds to 7.21 (7.211102…)
						Total 6 marks

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 9*

9	eg $12\sin60\left(=6\sqrt{3}=10.3(9...)\right)$ **or** $\sqrt{12^2-"6"^2}\left(=6\sqrt{3}=10.3(9...)\right)$ **or** (Area ADC =) $\frac{1}{2}\times12\times47\times\sin60\,(=244.2...)$		5	M1	for a method find the height of the trapezium **or** the area of triangle ADC The first two M1 marks can be awarded in either order
	eg $12\cos60(=6)$ **or** $\sqrt{12^2-\left("6\sqrt{3}"\right)^2}\,(=6)$			M1	(indep) for a method find the base of the triangle, condone missing brackets around $"6\sqrt{3}"$ The first two M1 marks can be awarded in either order
	eg (AB =) 47 – "6" – "6" (= 35)			M1	(dep on previous M1) for method to find the length of AB
	eg (Trapezium =) $\frac{1}{2}\times(47+"35")\times"10.3(9...)"$ **or** (Rectangle + 2 × Triangle =) $"35"\times"10.3(9...)"+2\times\frac{1}{2}\times"6"\times"10.3(9...)"$ **or** (Rectangle + 2 × Triangle =) $"35"\times"10.3(9...)"+2\times\frac{1}{2}\times"6"\times12\times\sin60$ **or** (Triangle ADC + Triangle ABC =) $"244.2..."+\frac{1}{2}\times12\times"35"\times\sin120$ oe eg $(47-"6")\times"10.3(9...)"$			M1	for a complete method There are other methods and marks should be awarded for a complete method that should give the correct area
	Working required	426		A1	(dep on M1) allow 420 – 427 from correct working
					Total 5 marks

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 14*

14	(a)	(0.5 ×) 9.3 × 14.7 × sin106 **or** (9.3 × cos 16) × 14.7 **or** (9.3 × sin 74) × 14.7		2	M1 for applying the area of a triangle formula using correct values (to find half of the area of the parallelogram) **or** for a correct method to find the area of the parallelogram
			131		A1 awrt 131
	(b)	(GE^2 =) $9.3^2+14.7^2-2\times9.3\times14.7\times\cos106$		3	M1 for the correct use of the cosine rule
		377(.9….) **or** 378 **or** 86.49 + 216.09 + 75.3… **or** 302.58 + 75.3….			M1 (dep on M1)for the correct order of operations
			19.4		A1 for 19.4 – 19.5
					Total 5 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 17

17	(area PQS =) $\frac{1}{2}\times 6.1\times 3.8\times \sin P = 9$ **or** (area $PQRS$ =) $6.1\times 3.8\times \sin P = 18$	$\frac{1}{2}\times 6.1\times SX = 9$ **or** $(SX =)\frac{9}{\frac{1}{2}\times 6.1}\ (=2.95...)$ **or** $6.1\times SX = 18$ **or** $(SX =) 18\div 6.1 (=2.95...)$		5	M1	correct equation for the area of the triangle or parallelogram **or** a calculation to find the height of the parallelogram (where X is the point vertically below S on PQ)
	eg (sin P =) $\frac{9}{\frac{1}{2}\times 6.1\times 3.8}\left(=0.776... \text{ or } \frac{900}{1159}\right)$ **or** (sin P =) $\frac{18}{6.1\times 3.8}\left(=0.776... \text{ or } \frac{900}{1159}\right)$	$(PX^2 =) 3.8^2 - "2.95..."^2 (=5.73...)$ **or** $(PX =)\sqrt{3.8^2 - "2.95..."^2}\ (=2.39...)$			M1	correct expression for sin P **OR** for start of Pythagoras method to find length of PX (where X is the point vertically below S on PQ)
	$(P =)$ sin^{-1}"0.776..." (= 50.9...)	$(QX =) 6.1 - \sqrt{"5.73"}\ (=3.70...)$ **or** $(QX =) 6.1 - "2.39" (=3.70...)$			M1	for complete method to find angle P **OR** for method to find length of QX
	$(QS^2 =) 3.8^2 + 6.1^2 - 2\times 3.8\times 6.1\times \cos("50.9")$ (= 22.4...) **or** $(QS =)\sqrt{3.8^2 + 6.1^2 - 2\times 3.8\times 6.1\times \cos("50.9")}$	$(QS^2 =) "2.95..."^2 + "3.70..."^2 (=22.4...)$ **or** $(QS =)\sqrt{"2.95..."^2 + "3.70..."^2}$			M1	correct expression for QS^2 (or QS)
			4.74		A1	accept 4.73 – 4.74
						Total 5 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 11

11	$(AX =)$ (17.6 – 8.4) ÷ 2 (= 4.6)		6	M1	where X is the foot of the perpendicular from B to AD
	0.5 × (8.4 + 17.6) × h = 179.4 **or** 0.5 × '4.6' × h + 0.5 × '4.6' × h + 8.4 × h = 179.4 **or** 13 × h = 179.4			M1	
	$(h =)$ 179.4 ÷ '13' (=13.8) **or** $(h =)$ 358.8 ÷ '26' (=13.8) oe			M1	
	tan $ABX = \frac{'4.6'}{'13.8'}$ **or** tan $BAX = \frac{'13.8'}{'4.6'}$			M1	ft their h dep on second M1 $(AB =)\sqrt{'4.6'^2 + '13.8'^2} = \sqrt{211.6}$ = (14.546…) **and** one from sin $ABX = \frac{'4.6'}{'\sqrt{211.6}'}$ or sin $BAX = \frac{'13.8'}{'\sqrt{211.6}'}$ or cos $ABX = \frac{'13.8'}{'\sqrt{211.6}'}$ or cos $BAX = \frac{'4.6'}{'\sqrt{211.6}'}$ or sin $ABX = \frac{'4.6'\times \sin 90}{'\sqrt{211.6}'}$ or cos $ABX = \frac{'\sqrt{211.6}'^2 + '13.8'^2 - '4.6'^2}{2\times '\sqrt{211.6}'\times '13.8'}$
	$(ABX =)$ tan$^{-1}\left(\frac{'4.6'}{'13.8'}\right)(=18.4)$ **or** $(BAX =)$ tan$^{-1}\left(\frac{'13.8'}{'4.6'}\right)(=71.6)$			M1	
		108.4		A1	awrt 108.4
					Total 6 marks

A+ Maths Hub International Community Schools